

NPTEL Week 12

ASSIGNMENT 12 SOLUTIONS & EXPLANATIONS

PROGRAMMING SOLUTIONS

1. PageRank Transfer

// PYTHON

```
rank_a, rank_b = [int(x) for x in input().split()]
print(0, rank_a + rank_b, end='')
```

// CONSOLE INJECTION

```
const el = document.querySelector('textarea.inputarea, textarea');
if (el) {
  el.focus();
  document.execCommand('selectAll', false, null);
  document.execCommand('insertText', false, `rank_a, rank_b = [int(x) for x in input().split()]
print(0, rank_a + rank_b, end='')`);
}
```

2. Mutual PageRank Exchange

// PYTHON

```
rank_a, rank_b = [int(x) for x in input().split()]
print(rank_b, rank_a, end='')
```

// CONSOLE INJECTION

```
const el = document.querySelector('textarea.inputarea, textarea');
if (el) {
  el.focus();
  document.execCommand('selectAll', false, null);
  document.execCommand('insertText', false, `rank_a, rank_b = [int(x) for x in input().split()]
print(rank_b, rank_a, end='')`);
}
```

3. Three-Page PageRank Update

// PYTHON

```
rank_a, rank_b, rank_c = [int(x) for x in input().split()]
print(0, rank_a // 2, rank_a // 2 + rank_b + rank_c, end='')
```

```
const el = document.querySelector('textarea.inputarea, textarea');
if (el) {
  el.focus();
  document.execCommand('selectAll', false, null);
  document.execCommand('insertText', false,
    `rank_a, rank_b, rank_c = [int(x) for x in input().split()]
print(0, rank_a // 2, rank_a // 2 + rank_b + rank_c, end='')`);
}
```

MCQ & MSQ SOLUTIONS WITH EXPLANATIONS

Scenario 1: Ranking the Most Influential Campus Clubs

Q01 Because it considers the importance of the pages providing the links.

Explanation: PageRank weighs both the quantity and the quality of incoming links. A link from an important page counts for more than a link from an obscure one.

Q02 The new club's PageRank is likely to improve due to the influential incoming link.

Explanation: A highly ranked page passes on a large share of its rank, so the new club gains. It won't automatically jump to the top, though.

Q03 Each outgoing link shares only a smaller portion of the page's ranking value.

Explanation: A page's rank is divided equally among its outgoing links. With hundreds of links, each one carries only a tiny share.

Q04 Importance is determined iteratively based on the network of incoming links.

Explanation: Ranks are recomputed over several rounds from the link structure until they settle. Content length, age and file size play no part.

Q05 Selected Options: MSQ

—A link from an influential page generally contributes more than a link from an unimportant page.

—The ranking values are refined over multiple iterations.

Explanation: PageRank depends on link quality, not just the count, and it converges over iterations. Outgoing links pass on rank/out-degree, not zero.

Scenario 2: Ranking Webpages Using a Simplified PageRank Score

Q06 It divides the page's current score equally among its outgoing links.

Explanation: $\text{share} = \text{rank}[\text{page}] / \text{len}(\text{links}[\text{page}])$ splits the page's score evenly among the pages it links to.

Q07 C

Explanation: Links: $A \rightarrow [B, C]$, $B \rightarrow [C]$, $C \rightarrow [A, B, D]$, $D \rightarrow [A]$. Totals: $A = 1/3 + 1 = 1.33$, $B = 0.5 + 1/3 = 0.83$, **$C = 0.5 + 1 = 1.5$** , $D = 1/3 = 0.33$.

Q08 C

Explanation: C has the most outgoing links (3), so each linked page gets only 1/3. A gives 0.5 per link, and B and D each give 1.

Q09 new_rank

Explanation: Contributions are added into `new_rank[dest]`, and line 17 prints `new_rank`. `rank` keeps the original values.

Q10 Selected Options: MSQ

—Every page starts with the same initial PageRank value.

—The program performs exactly one PageRank iteration.

Explanation: All pages start at 1.0, and the loop over the pages runs once. More outgoing links means a *smaller* share per link, and `rank` is never modified.

Scenario 3: Evaluating One Iteration of a Simplified PageRank Algorithm

Q11 R

Explanation: R links only to S, so it passes its entire 1.0 to S. P gives 0.5 per link, Q gives 1/3 and S gives 0.5.

Q12 Q

Explanation: $P \rightarrow [Q, R]$ and $S \rightarrow [P, Q]$. Q is the only page that both P and S link to.

Q13 S

Explanation: $P = 1/3 + 0.5 = 0.83$, $Q = 0.5 + 0.5 = 1.0$, $R = 0.5 + 1/3 = 0.83$, **$S = 1/3 + 1 = 1.33$** .

Q14 To ensure every contribution is computed using the original ranks.

Explanation: If `rank` were updated mid-loop, pages processed later would distribute their new values instead of their original ones, which would give wrong results.

Q15 Selected Options: MSQ

—A page with fewer outgoing links transfers a larger share to each linked page.

—The total contribution distributed by each page is equal to its current rank.

—All updated ranks are computed before replacing the original ranks.

Explanation: $\text{Share} = \text{rank} / \text{out-degree}$, so fewer links means a bigger share, and the shares add back up to the page's rank. All values go into `updated` while `rank` stays unchanged during the loop.

Scenario 4: Exploring the Collatz Conjecture

Q16 3

Explanation: $8 \rightarrow 4 \rightarrow 2 \rightarrow 1$. The loop runs 3 times, so `steps += 1` (line 10) executes 3 times.

Q17 It checks whether the current number is even.

Explanation: `if n % 2 == 0:` is true when `n` is divisible by 2.

Q18 5

Explanation: Input 3 prints 3, then becomes 10. 10 is printed, and because it is even it is halved to 5. 16 would come from wrongly applying $3n+1$ to 10.

Q19 It counts the number of transformations performed before reaching 1.

Explanation: steps increases once per loop iteration, and each iteration applies one transformation.

Q20 Selected Options: MSQ

—The value 1 is printed even though the loop terminates before processing it.

—The variable steps is incremented exactly once during every loop iteration.

—The program prints every intermediate value generated by the sequence.

Explanation: while $n \neq 1$ stops before printing 1, so `print(1)` on line 12 prints it. For odd n , $3n+1$ is always *larger*, so that option is false.
